CCS2512 Computer Graphics

INTRODUCTION

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Course Overview

- Credit Value: 2
- Core
- Hourly Breakdown:
  - Theory - 30 hrs
  - Practical - 30 hrs
  - Independent Learning - 40 hrs

What you will learn

- Fundamentals of computer graphics algorithms
  - Will give a pretty good idea of how to implement lots of the things just shown
- We will concentrate on 3D, not 2D illustration or image processing
- Basics of real-time rendering and graphics hardware
- Basic OpenGL
  - Not the focus, though: Means, not the end.
- You will get C++ programming experience

What you will NOT learn

- OpenGL and DirectX hacks
  - Most become obsolete every 18 months anyway!
  - Does not really matter either: Graphics is becoming all software again (OpenCL, Larrabee, etc.)
- Software packages
  - CAD-CAM, 3D Studio MAX, Maya
  - Photoshop and other painting tools
- Artistic skills
- Game design

How much Math?

- Lots of simple linear algebra
  - Get it right, it will help you a lot!
- Some more advanced concepts
  - Homogeneous coordinates
  - Ordinary differential equations (ODEs) and their numerical solution
  - Sampling, antialiasing (some gentle Fourier analysis)
  - Monte-Carlo integration
- Always in a concrete and visual context

Beyond computer graphics

- Many of the mathematical and algorithmic tools are useful in other engineering and scientific context
- Linear algebra
- Splines
- Differential equations
- Monte-Carlo integration
- …

Assessment Strategy

Group Project – 30%

Final Written Examination (Open Book) – 70%

References

- Foley, Van Dam, Feiner, Hughes Computer Graphics Principles and Practice Second Edition
- S. Marschner, and P. Shirley, Fundamentals of Computer Graphics, CRC Press, 4th Ed., 2015.
- S. Guha, Computer Graphics Through OpenGL: From Theory to Experiments 3rd Edition, CRC, 2019.
- D.D. Hearn, M.P. Baker, and W. Carithers, Computer Graphics with OpenGL, 4th Ed., 2010

What are the applications of graphics?

Movies/special effects

More than you would expect

http://vimeo.com/9553622

https://www.youtube.com/embed/MnQLjZSX7xM

Video Games








Simulation

CAD-CAM & Design

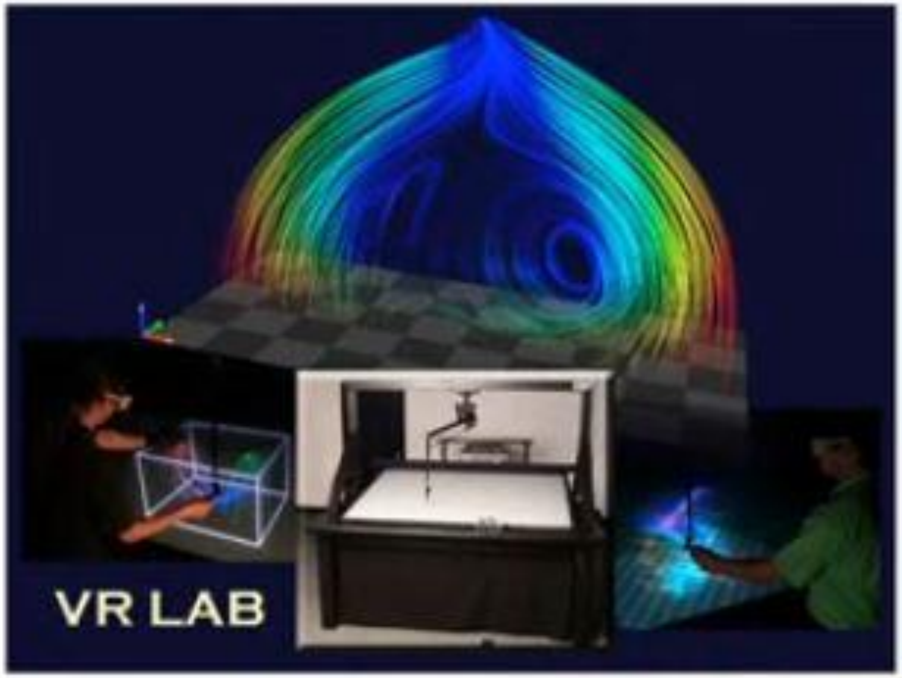


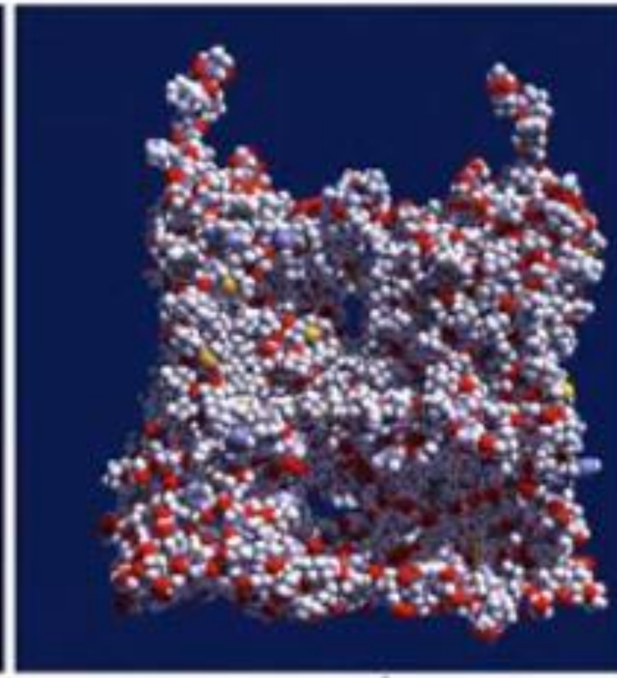

Architecture

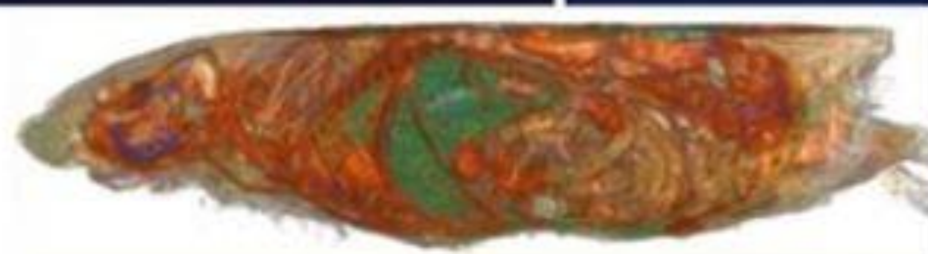
Virtual Reality

https://youtu.be/1IWfYS9DEbM?si=__kNcfFG64av5Gmu

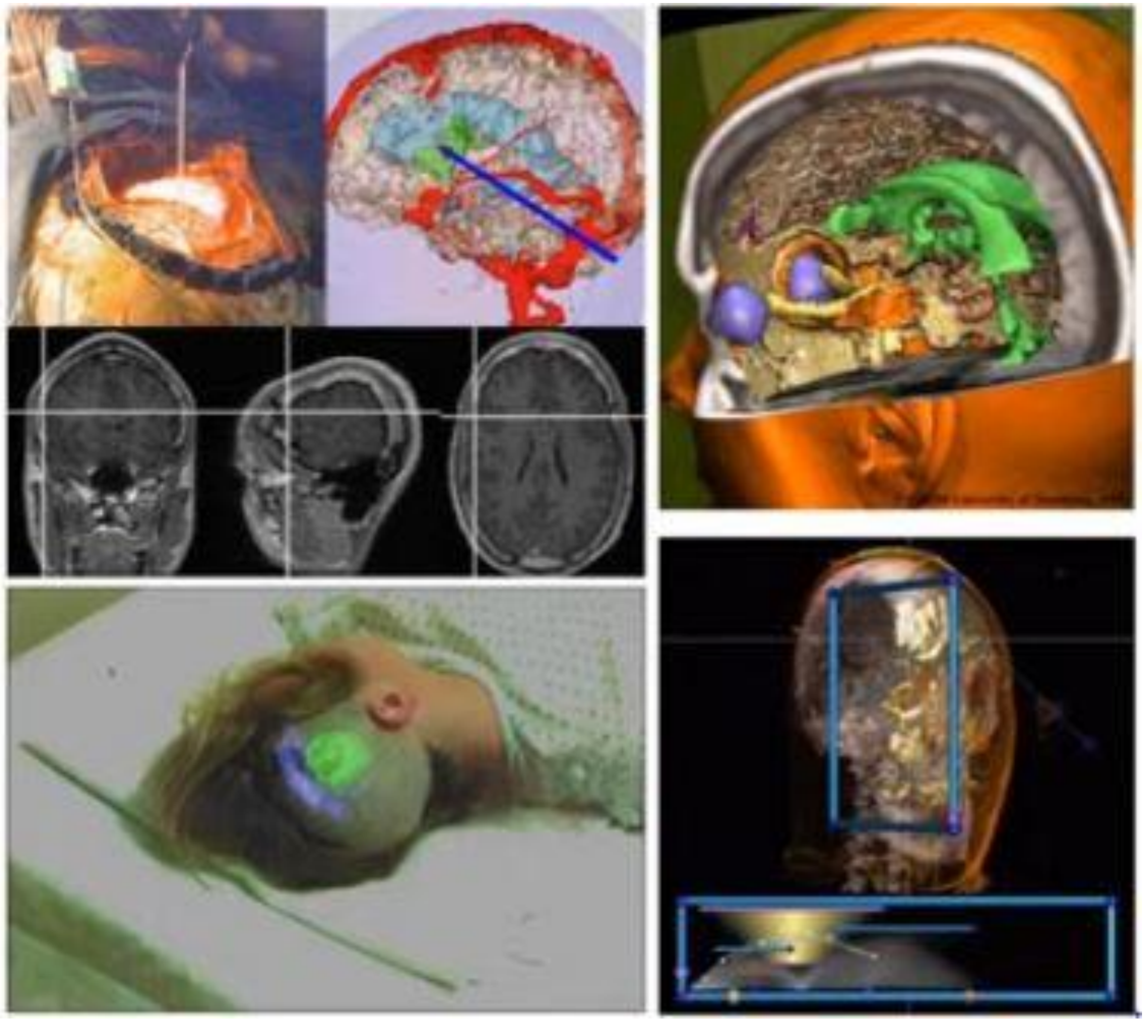
Visualization



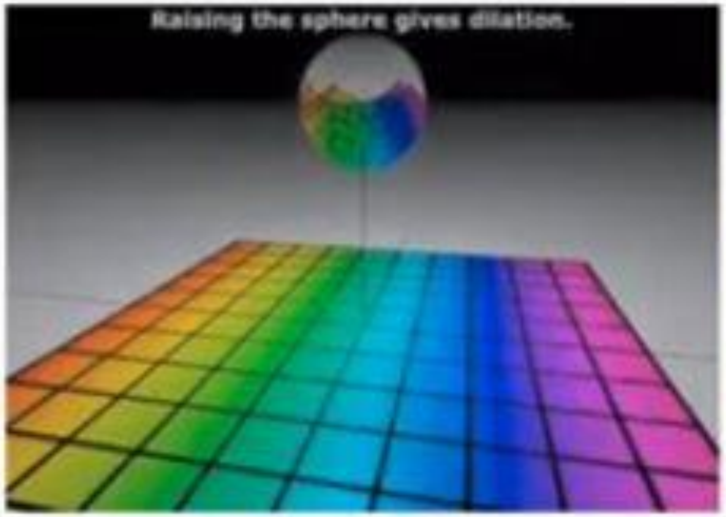


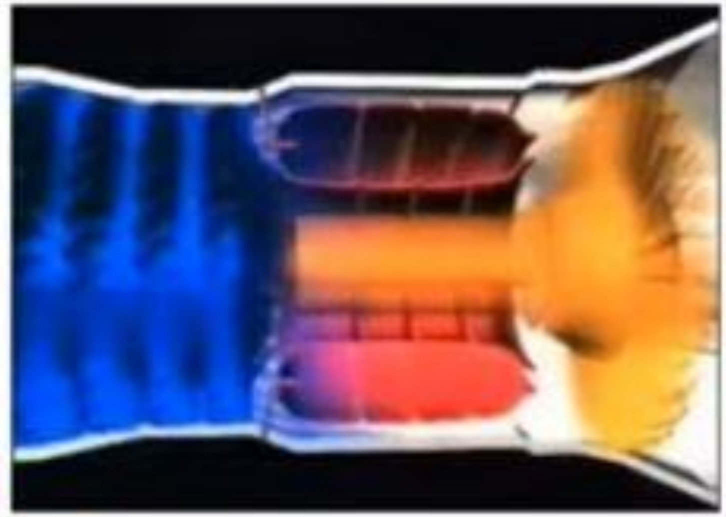


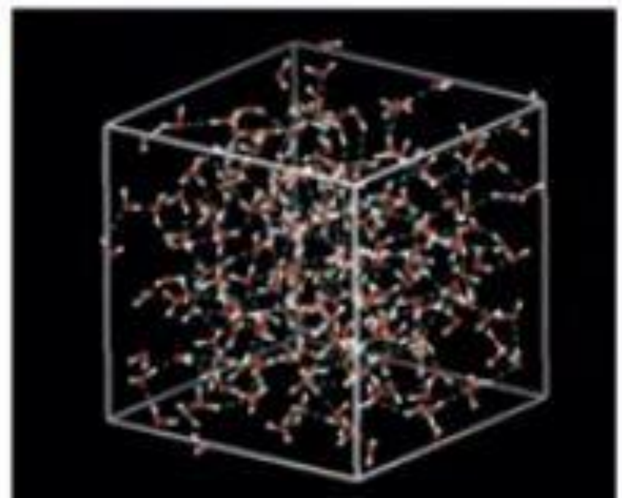
Medical Imaging



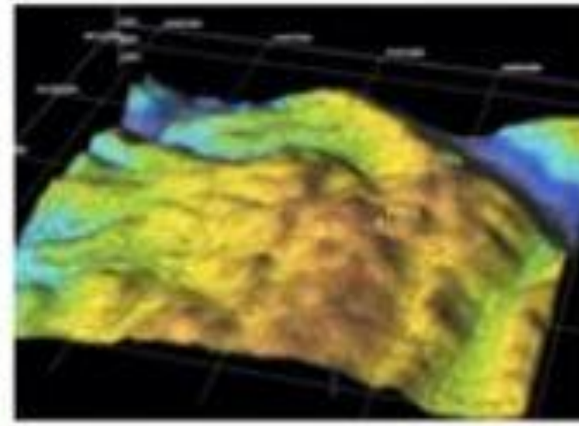
Education

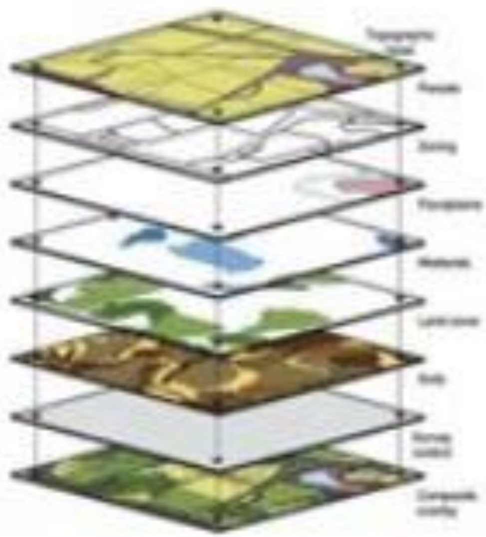






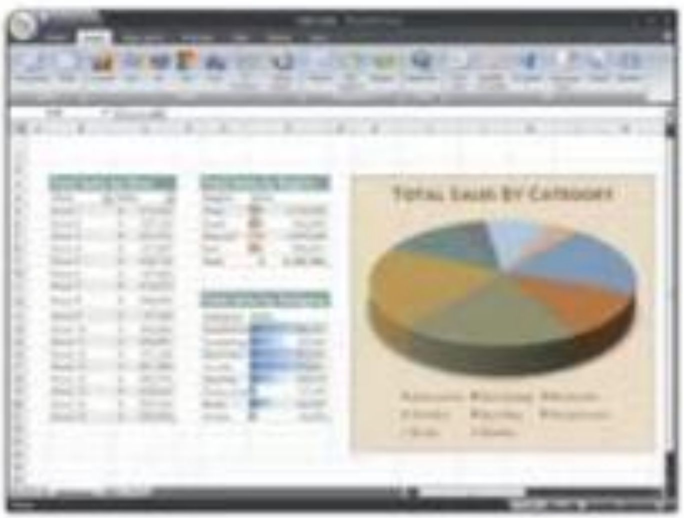
Geographic Info Systems & GPS





Any display

- Computers go through OpenGL and DirectX to display anything
- 2D graphics, Illustrator, Flash, Fonts



How do you make this picture?

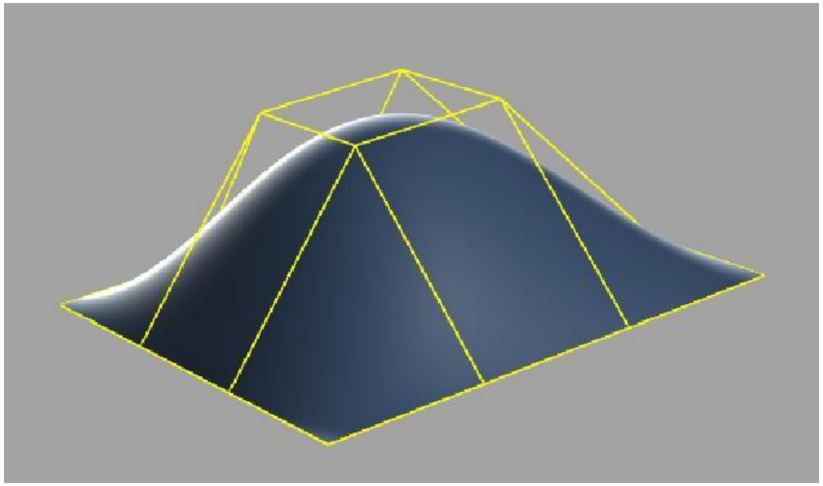
- Modeling
  - Geometry
  - Materials
  - Lights
- Animation
  - Make it move
- Rendering
  - I.e., draw the picture!
  - Lighting, shadows, textures…

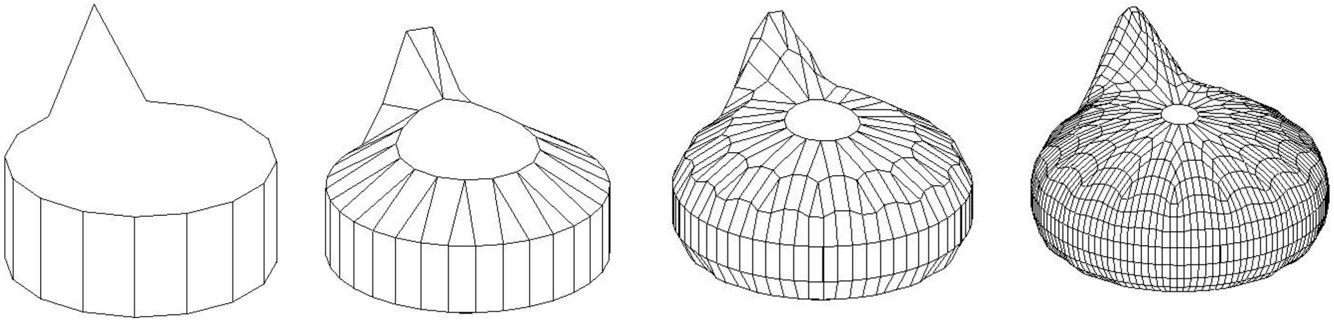
Overview of the Semester

- Modeling, Transformations
- Animation, Color
- Ray Casting / Ray Tracing
- The Graphics Pipeline
- Textures, Shadows
- Sampling, Global Illumination

Modeling

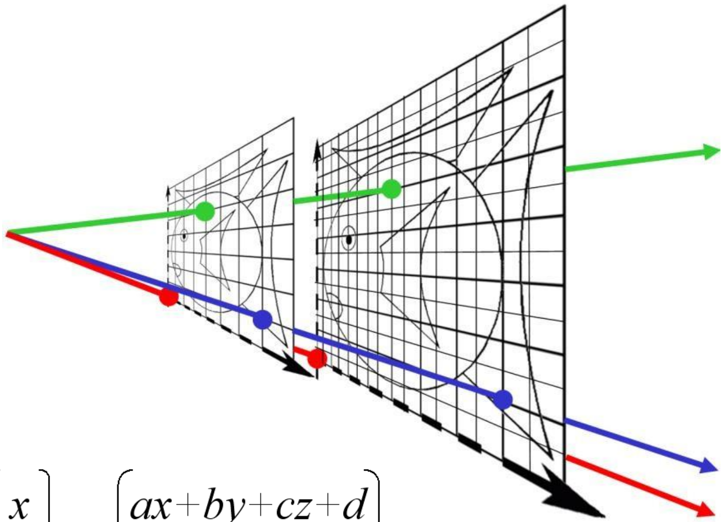
- Curves and surfaces
- Subdivision surfaces





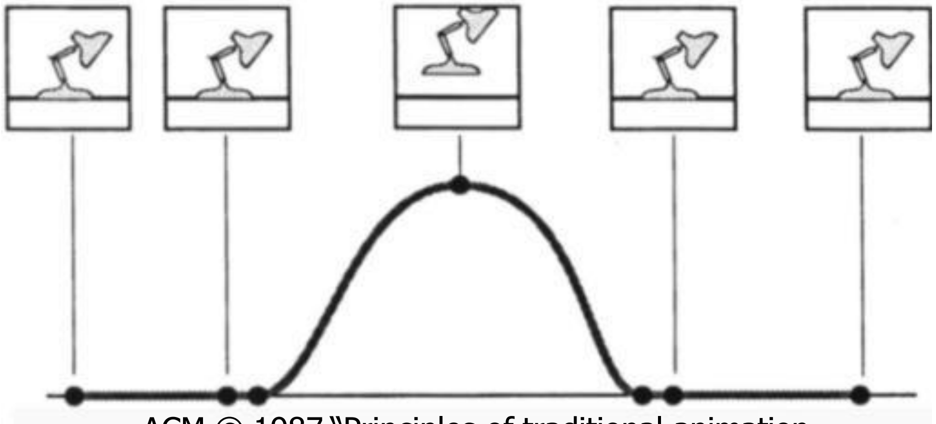
Transformations

- Yep, good old linear algebra
- Homogeneous coordinates
  - (Adding dimensions to make life harder)
- Perspective



$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} ax+by+cz+d \\ ex+fy+gz+h \\ ix+jy+kz+l \\ 1 \end{pmatrix}$$

Animation: Keyframing



ACM © 1987 "Principles of traditional animation applied to 3D computer animation"

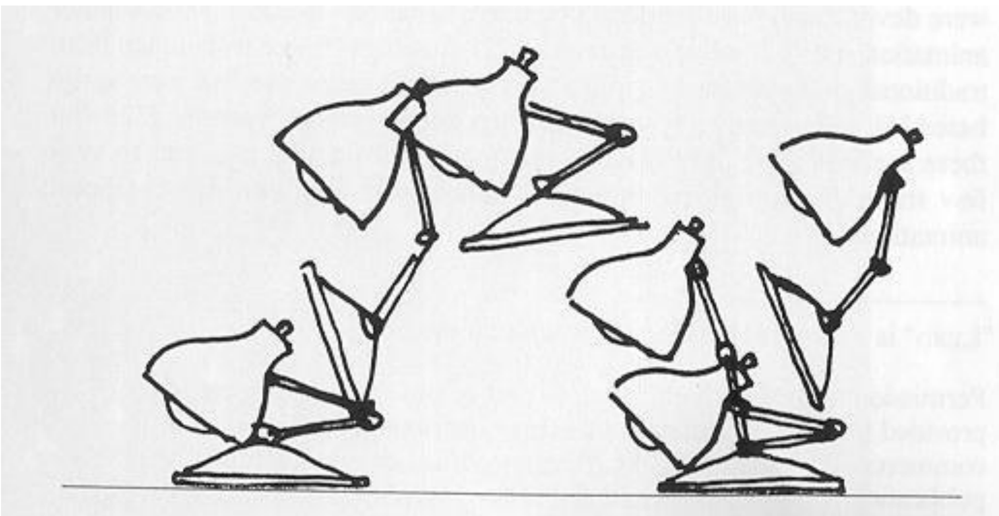
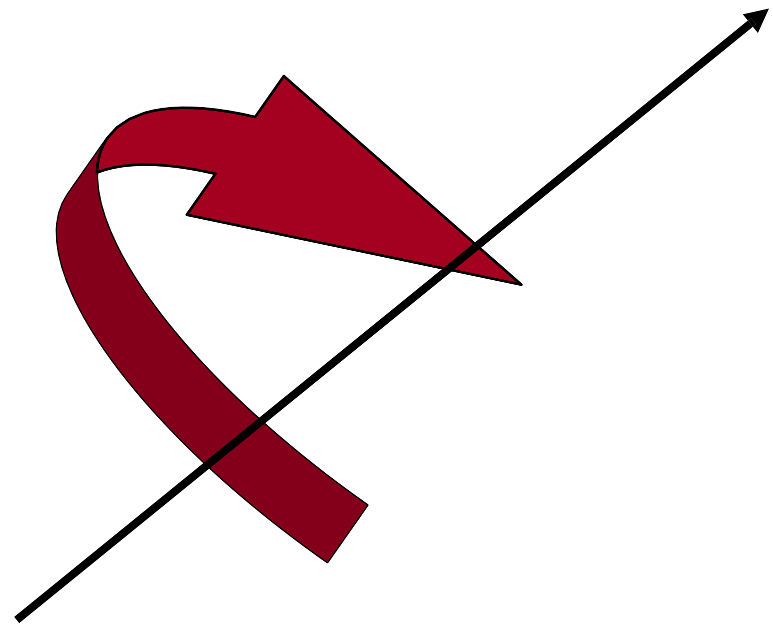
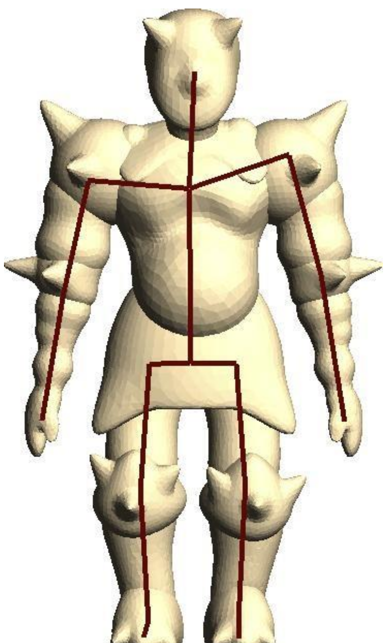
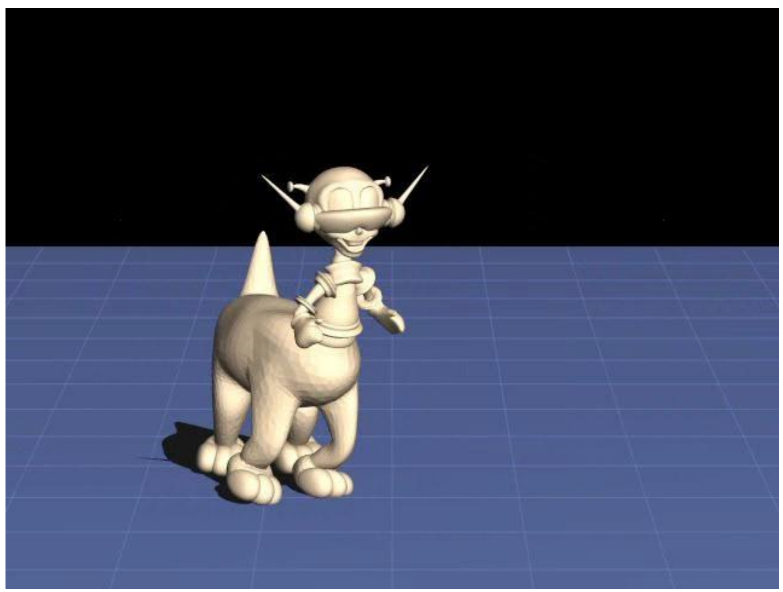


FIGURE 3. Squash & stretch in Luxo Jr.'s hop.



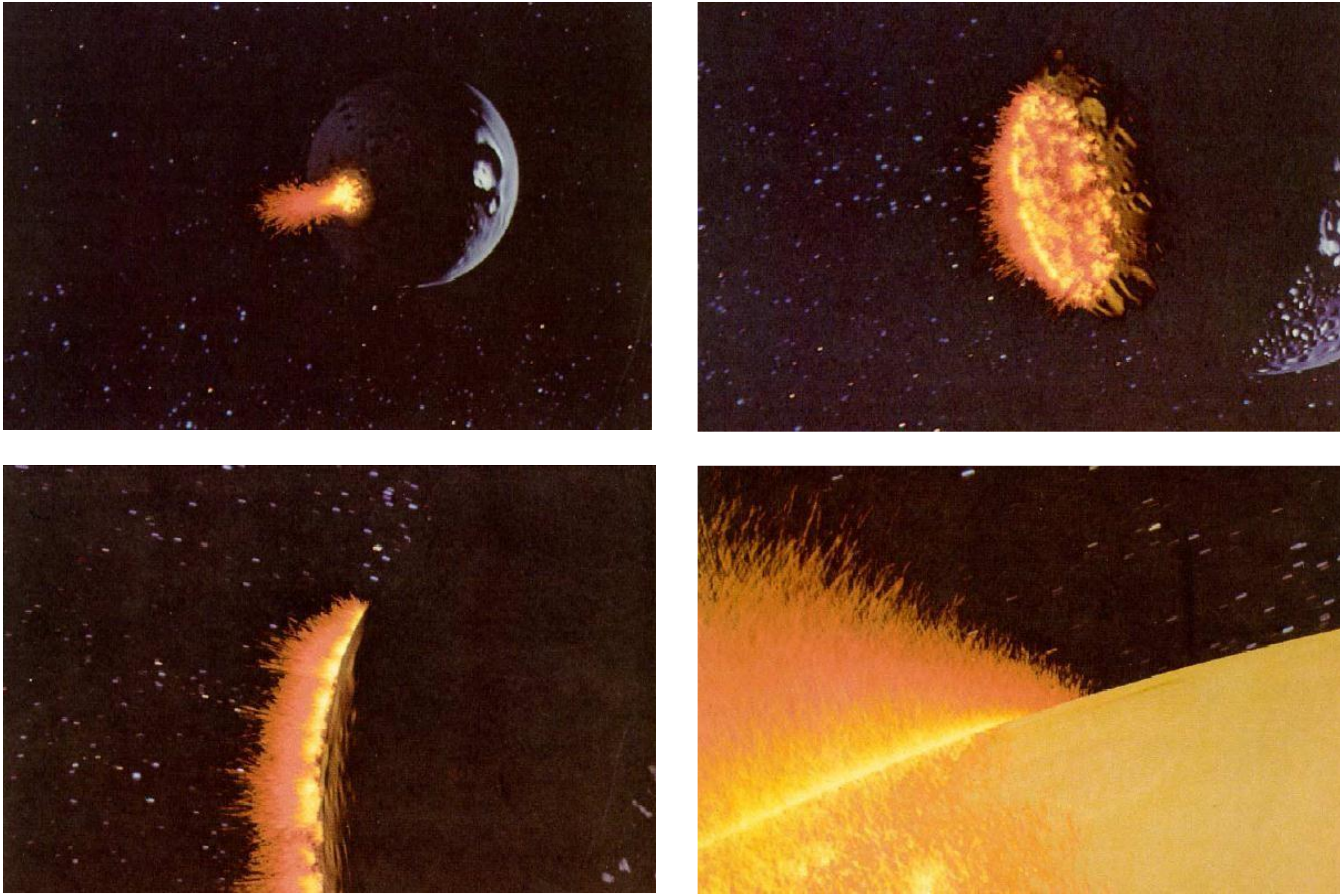
Character Animation: Skinning

- Animate simple “skeleton”
- Attach “skin” to skeleton
  - Skin deforms smoothly with skeleton
- Used everywhere (games, movies)

Ilya Baran

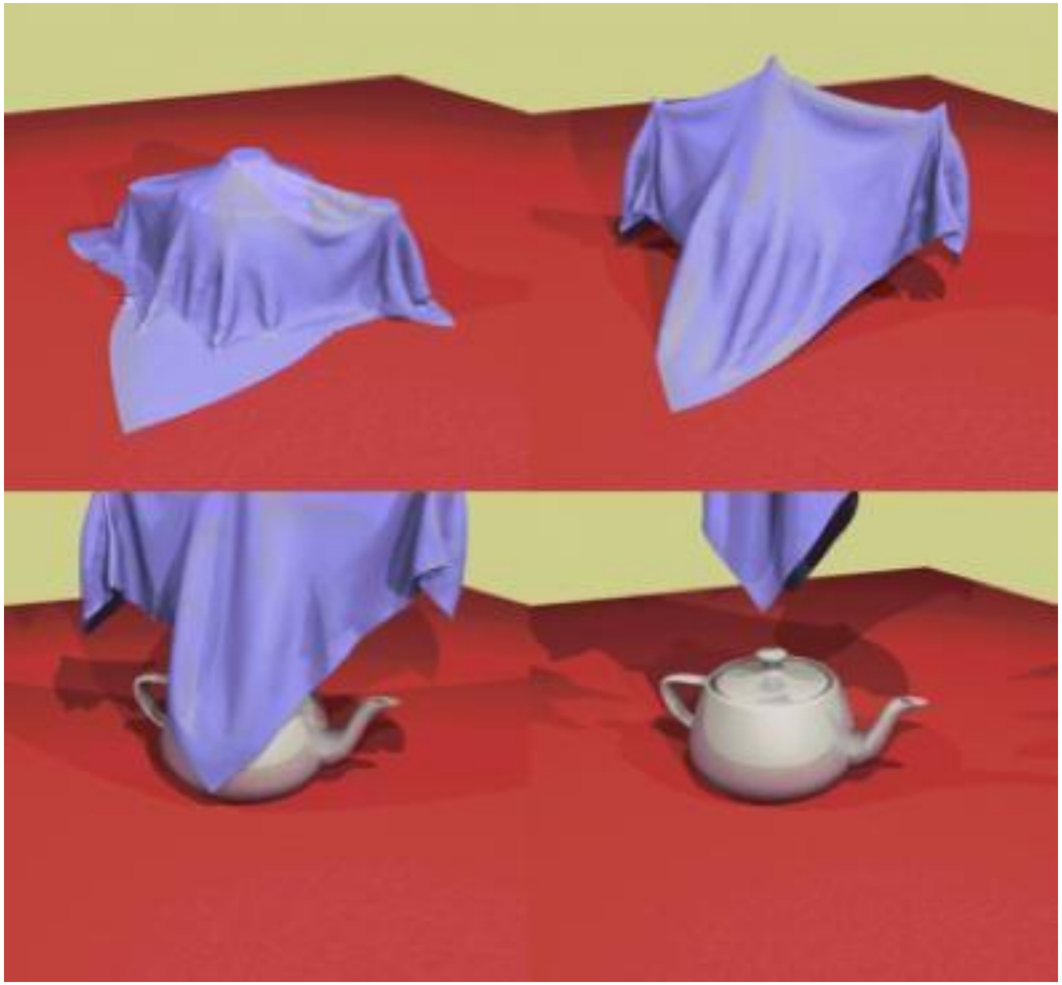
Particle system (PDE)



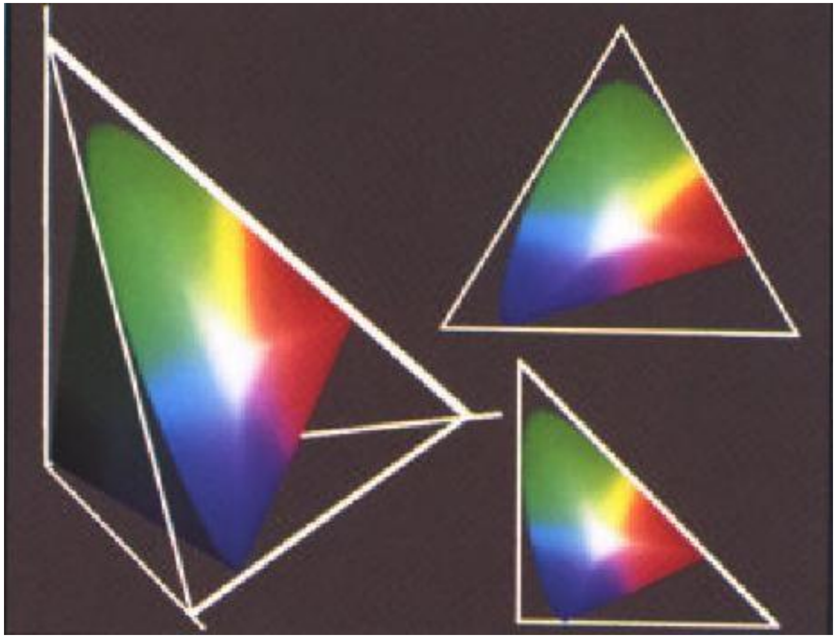
“Physics” (ODEs)

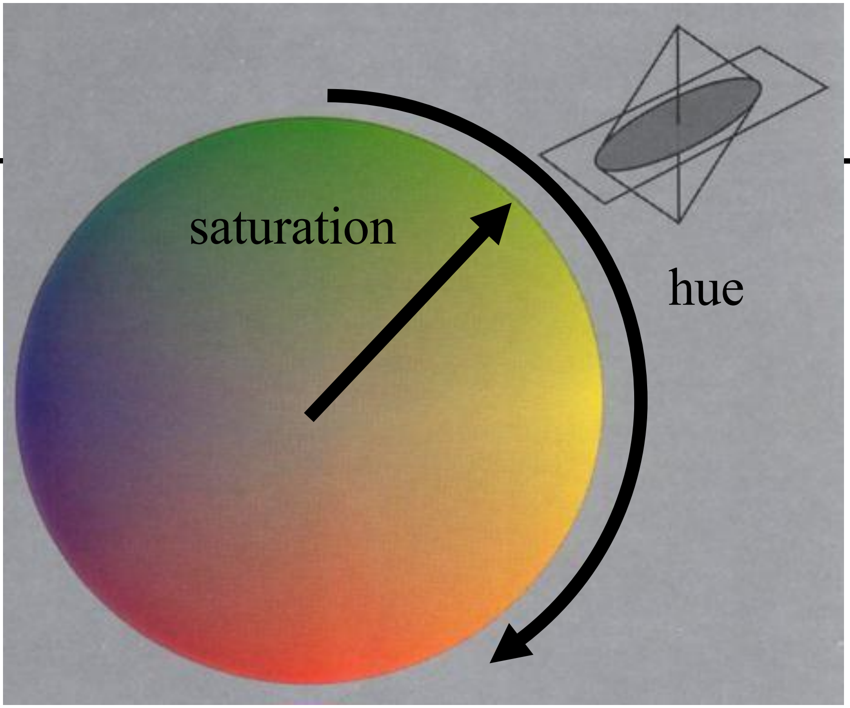
- Fire, smoke
- Cloth

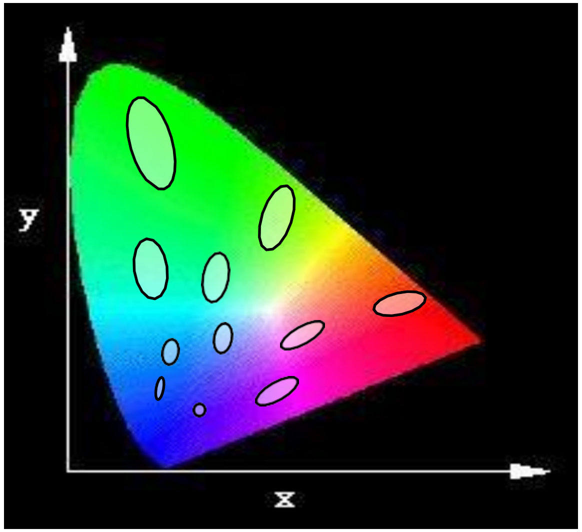
- Quotes because we do “visual simulation”

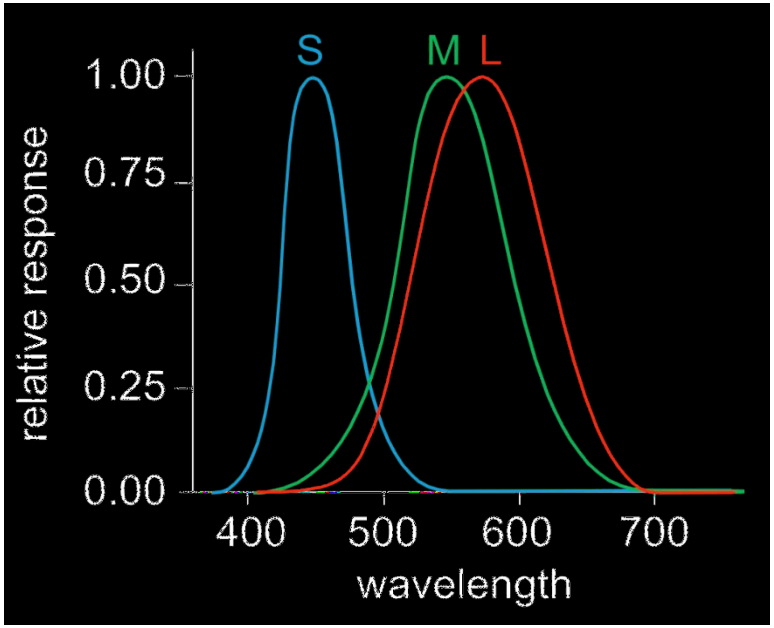


Color





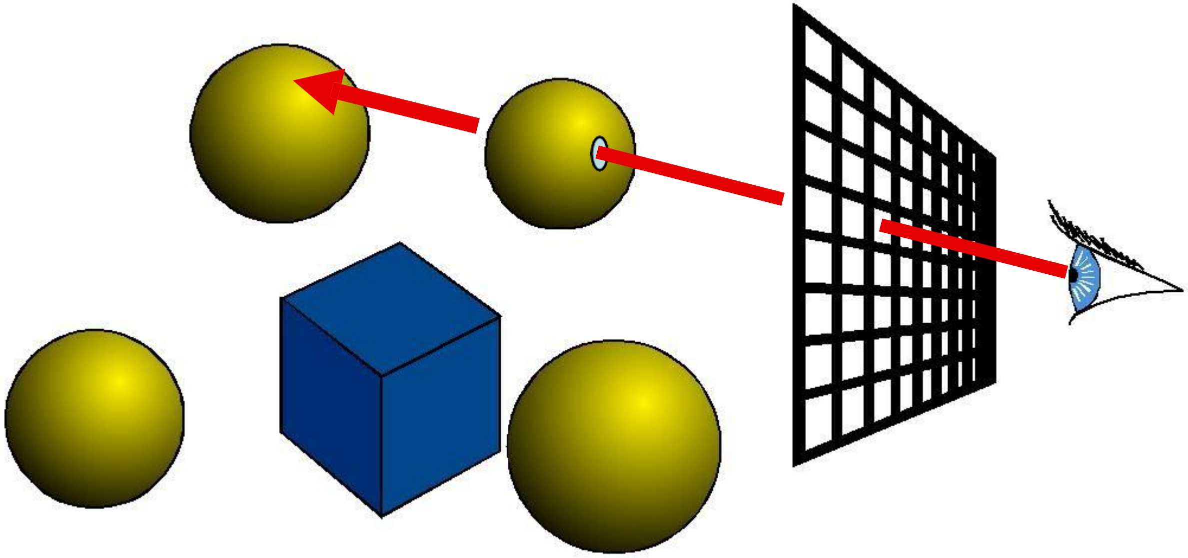




Courtesy of Victor Ostromoukhov.

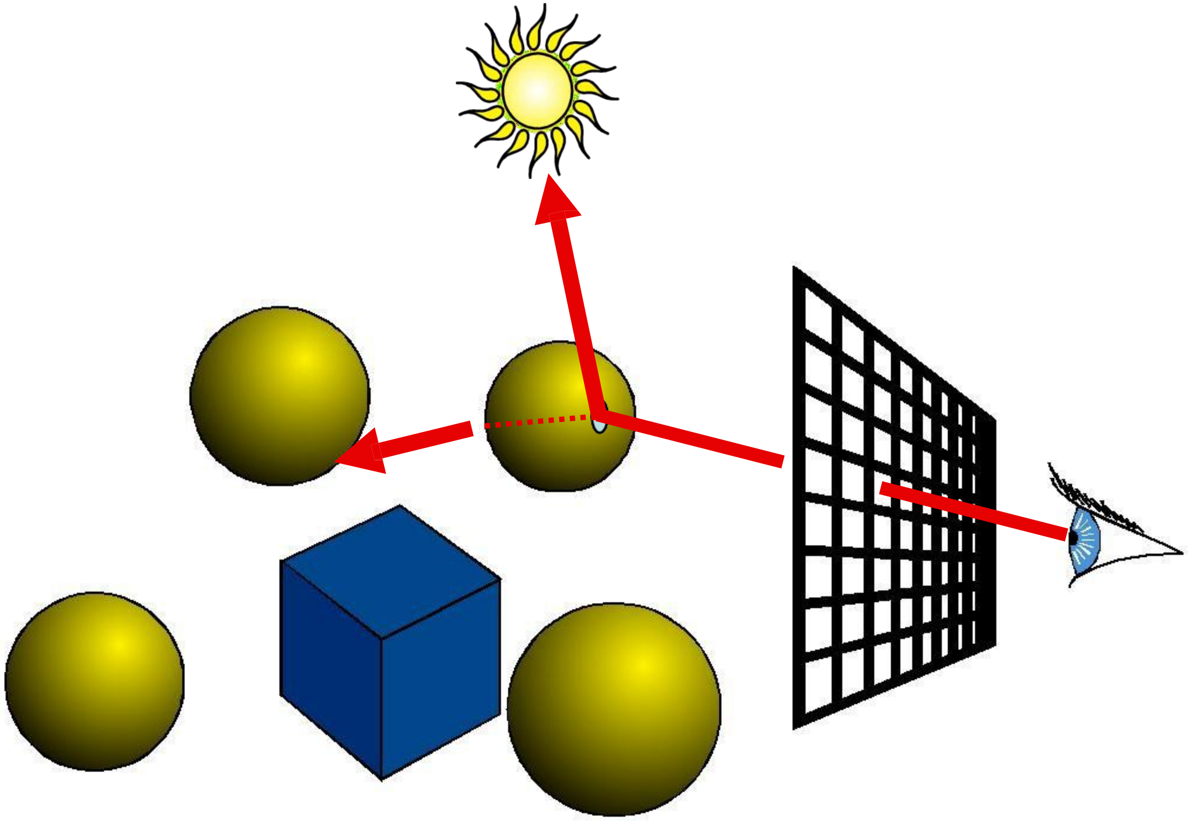
Ray Casting

- For every pixel
  construct a ray from the eye
  – For every object in the scene
    - Find intersection with the ray
    - Keep if closest



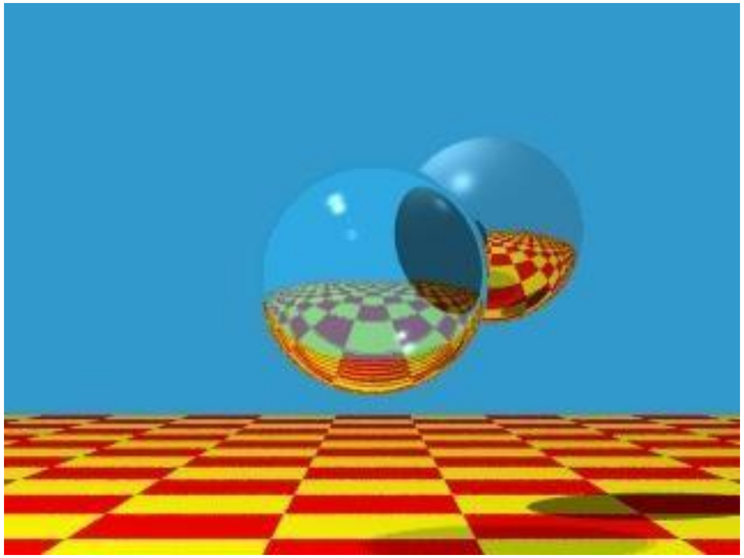
Ray Tracing

- Shade (interaction of light and material)
- Secondary rays (shadows, reflection, refraction)



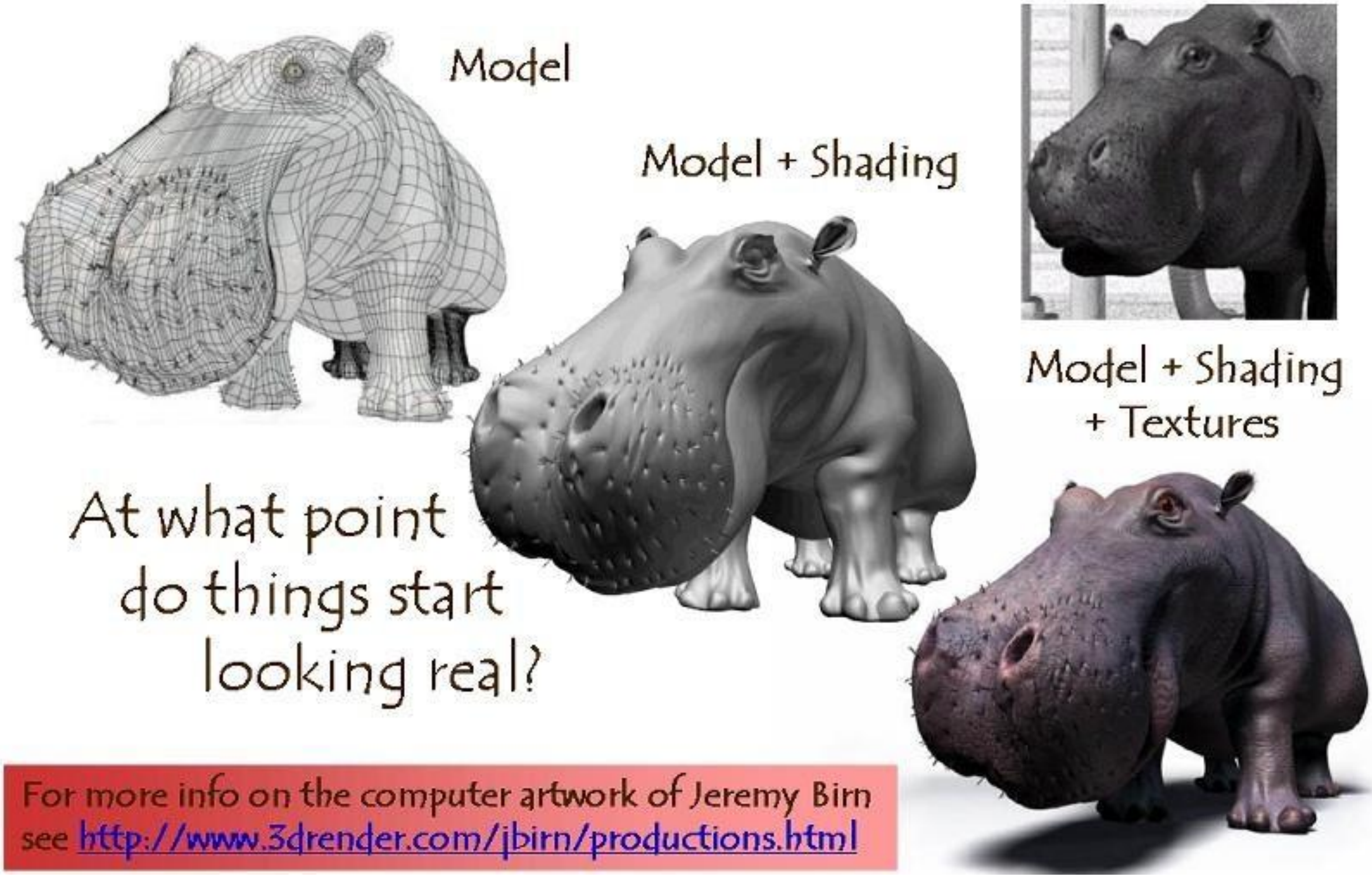
Ray Tracing

- Original Ray-traced image by Whitted



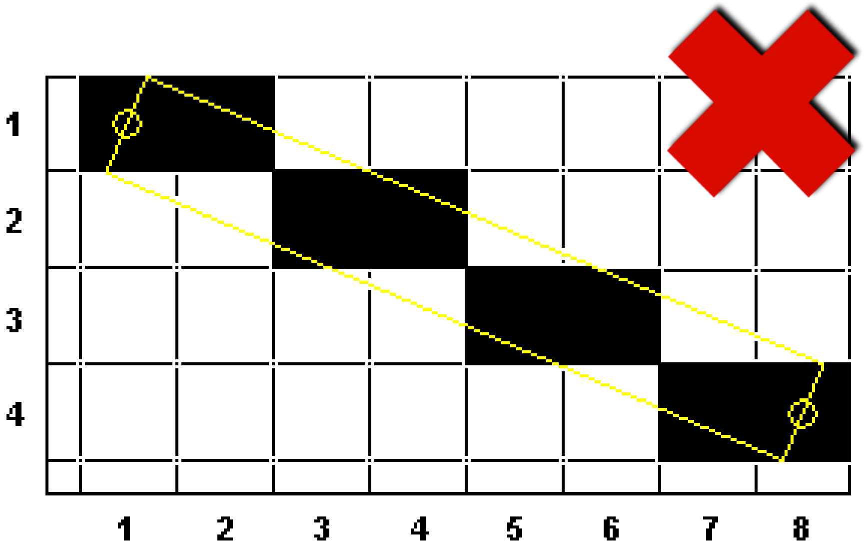
- Image computed using the Dali ray tracer by Henrik Wann Jensen
- Environment map by Paul Debevec

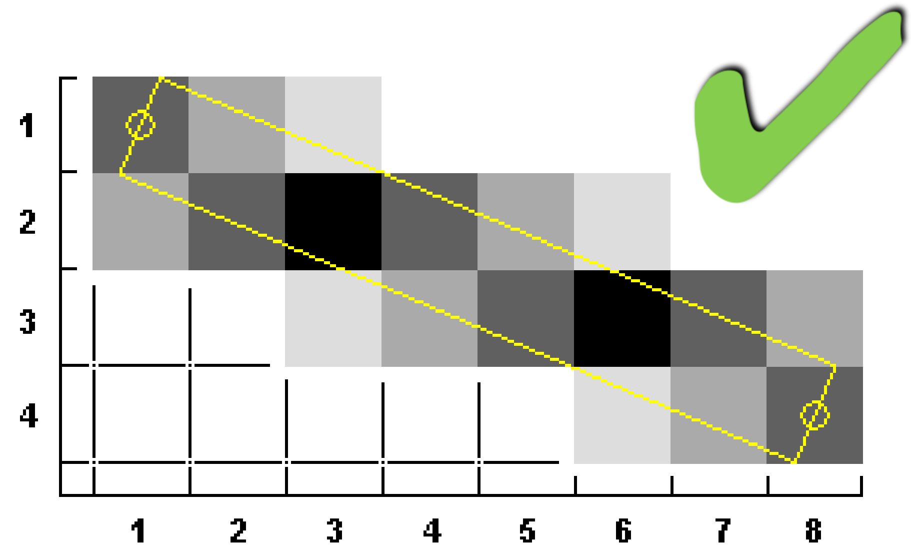
Textures and Shading

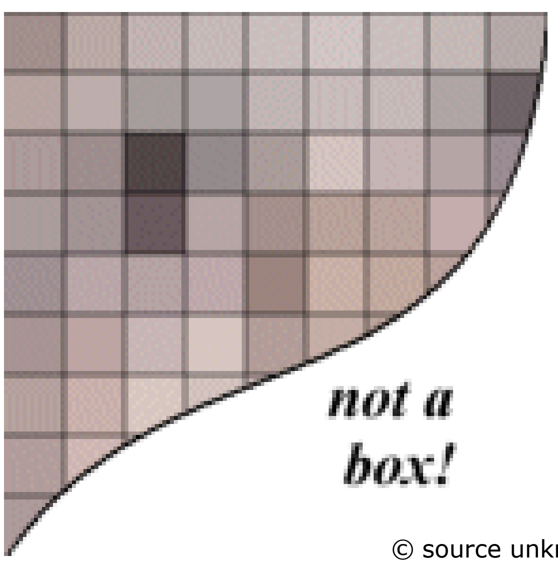


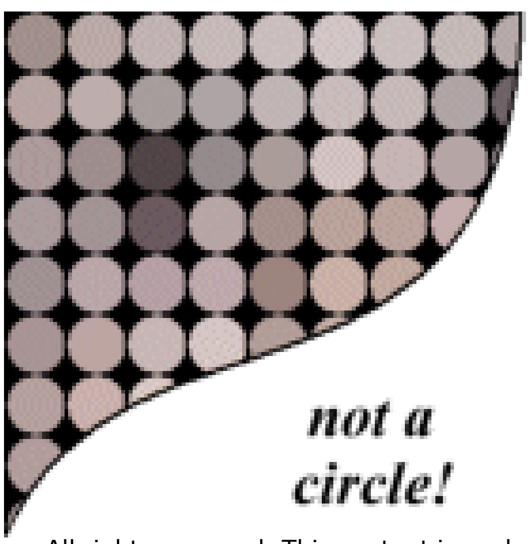
Courtesy of Jeremy Birn.

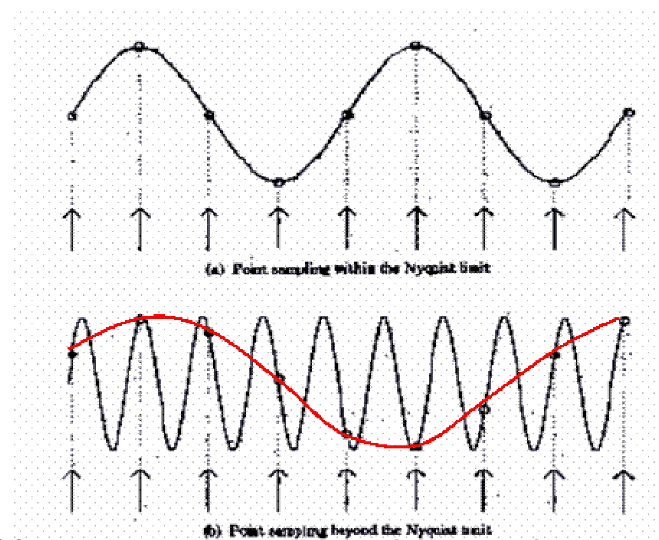
Sampling & Antialiasing











Shadows

Figure 12. Frame from *Luxo Jr.*

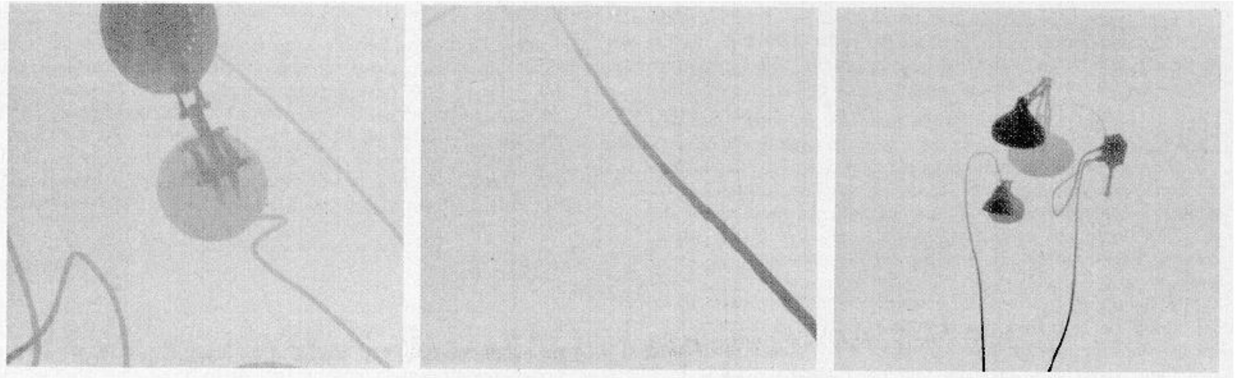
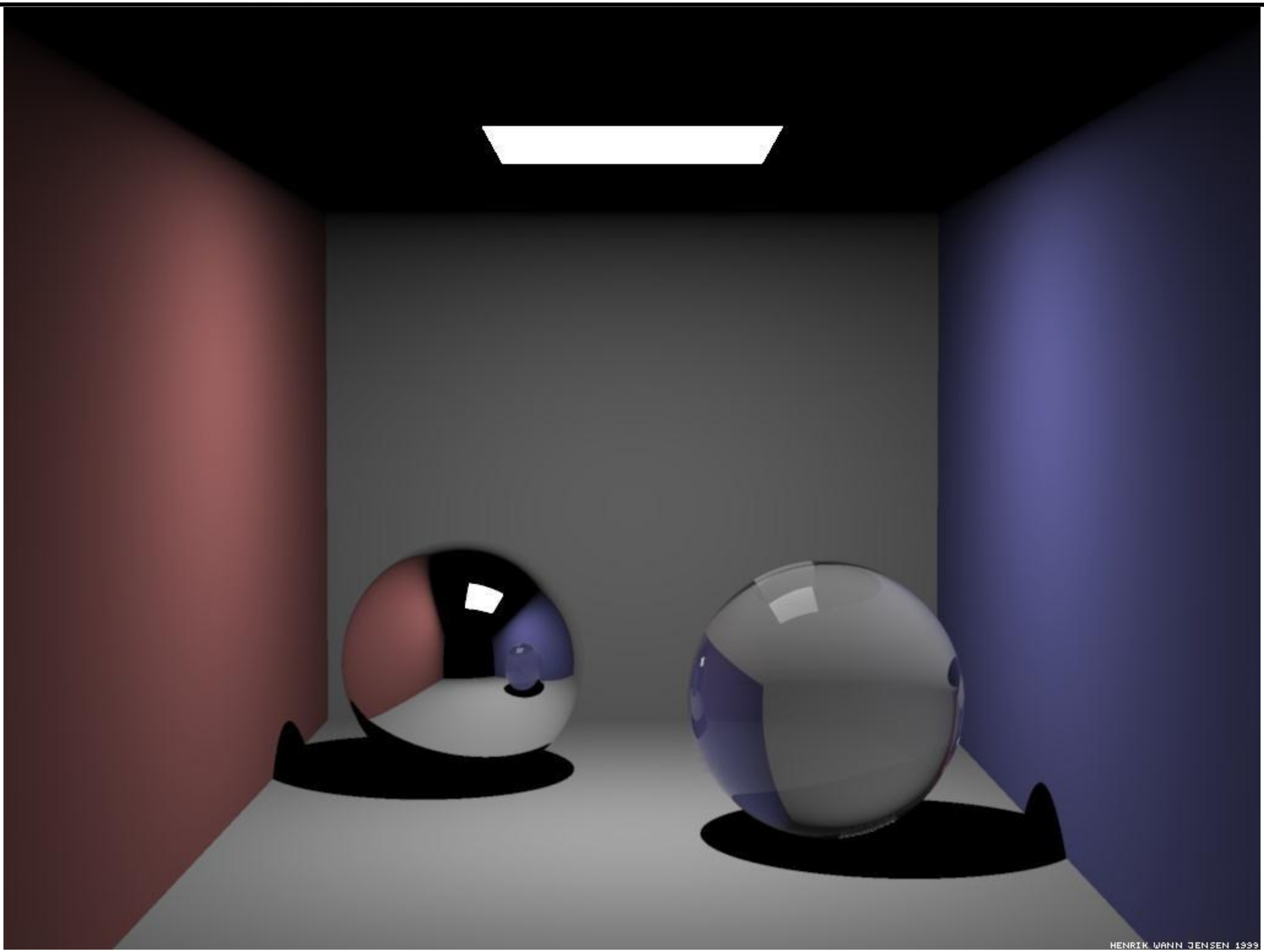
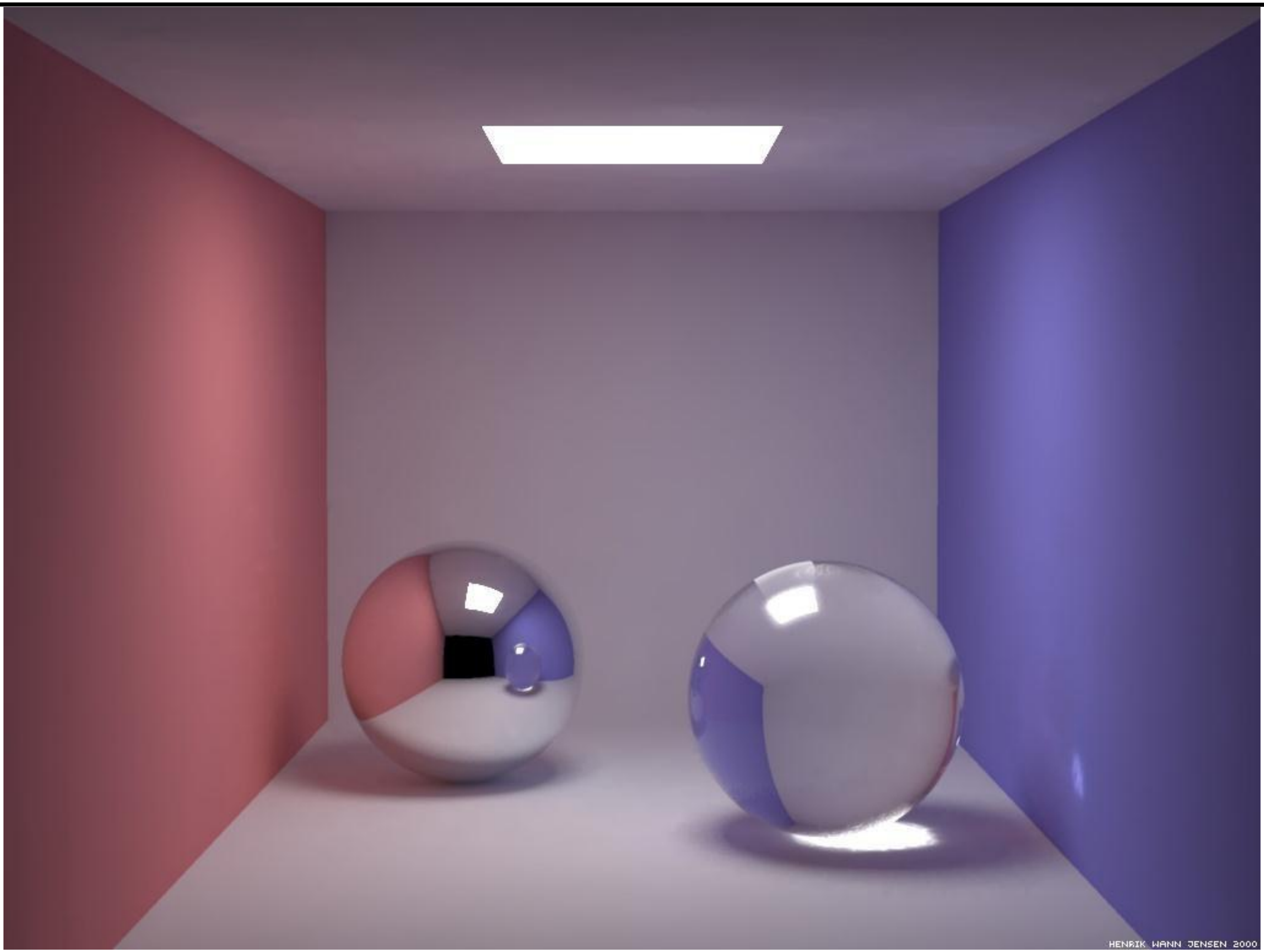


Figure 13. Shadow maps from *Luxo Jr.*

Traditional Ray Tracing



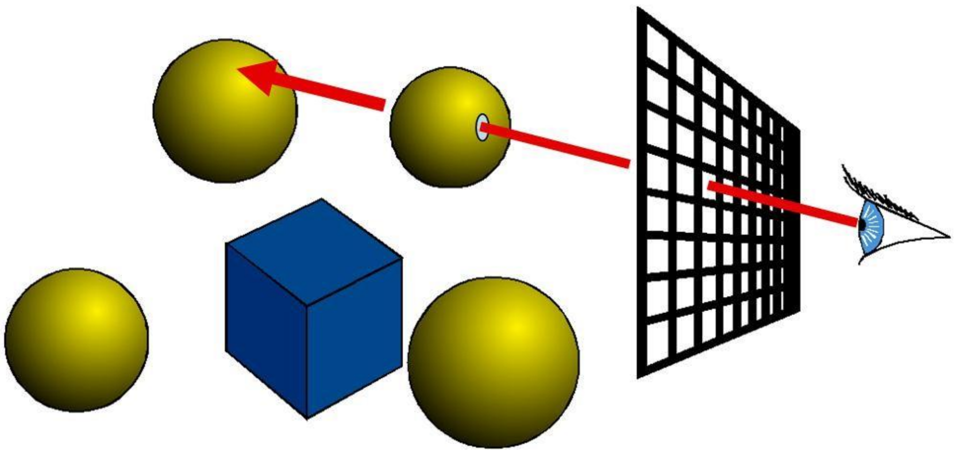
Global Illumination



The Graphics Pipeline

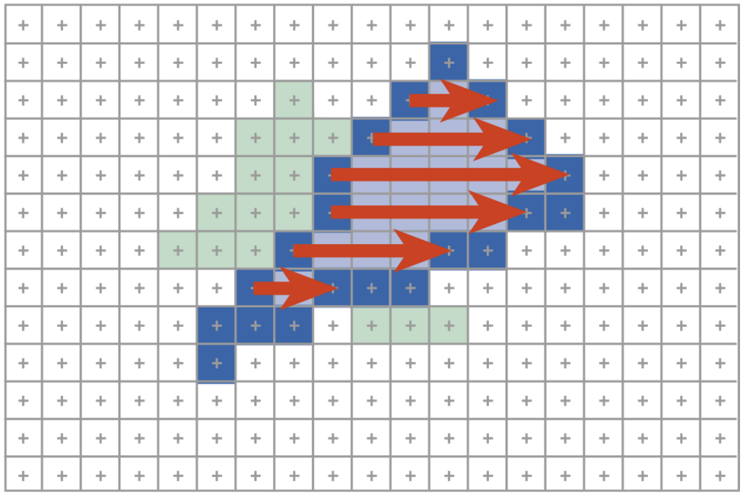
Ray Casting

```
For each pixel
    For each object
        Send pixels to scene
```



Rendering Pipeline

```
For each triangle
    For each projected pixel
        Project scene to pixels
```



The Graphics Pipeline

- Transformations
- Clipping
- Rasterization
- Visibility